A Hadamard matrix with growth factor greater than its order

Matias Andia*

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Abstract

It has long been conjectured that the growth factor of a completely pivoted Hadamard matrix is equal to its order. In this paper, we construct a row and column permutation of the Sylvester Hadamard matrix of order 4096 that is completely pivoted and has growth factor $4608 = (9/8)4096$, disproving the conjecture. This construction can be extended to give a completely pivoted Hadamard matrix of order 2^d with growth factor $(9/8)2^d$ for every integer $d \geq 12$.

1 Introduction

A matrix $H \in \mathbb{R}^{n\times n}$ is a Hadamard matrix of order n if its entries are ± 1 and $HH^\top = nI$. We say that H is *completely pivoted* (CP) if, at every step of Gaussian elimination without row or column interchanges, the pivot has maximum magnitude among the entries of the current Schur complement. We write p_k for the magnitude of the k-th pivot. The growth factor is the largest entry magnitude encountered during elimination, divided by the largest entry magnitude of the original matrix. For a CP Hadamard matrix, it is therefore

$$g(H) = \max_{1\leq k\leq n} p_k.$$

For a CP Hadamard matrix H of order n, the final pivot has magnitude n, so $g(H) \geq n$. A long-standing conjecture asserts that equality always holds [1, 2]. Equivalently, the conjecture states that no earlier pivot can exceed the final one in magnitude. Much of the previous work on this question has focused on the possible pivot sequences of CP Hadamard matrices (see, e.g., Day and Peterson [2] and Edelman and Friedman [3]). For a broader discussion of growth factors under complete pivoting, see Edelman and Urschel [4].

We construct a counterexample by permuting the rows and columns of a Sylvester Hadamard matrix. For $n = 2^d$, let W_n be the Sylvester matrix, defined by

$$W_1 = (1), \qquad W_{2n} = \begin{pmatrix} W_n & W_n \\ W_n & -W_n \end{pmatrix}.$$

Theorem 1. *There are permutation matrices P, Q for which $H = PW_{4096}Q$ is completely pivoted and*

$$g(H) = p_{4089} = 4608 = \frac{9}{8}4096.$$

Our permutations allow us to reach a 14×14 Schur complement with entries $\pm n/4$. Six further eliminations produce an entry of magnitude $9n/8$. The matrix is shown in Figure 2.

2 Construction and verification

We now describe the construction of the completely pivoted ordering in Theorem 1. The main idea is to begin with a short pivot sequence on a small Sylvester matrix, lift it to a larger matrix, and continue elimination on the remaining Schur complement. We repeat this procedure until the desired pivot appears.

*Department of Mathematics, Princeton University, Princeton, NJ. mati@princeton.edu

Lifting a pivot sequence. The following lemma formalizes the lifting step described above. It is a version of an observation due to Tornheim and is given in Peca-Medlin [6, Proposition 2.10]. We include a proof for completeness.

Lemma 1. *Suppose S is a matrix whose upper-left entry $p = S_{11} \neq 0$ is a complete pivot, so that $|p| = \|S\|_{\max}$. Then the first two pivots of $S \otimes W_2$ are complete pivots, with values p and $-2p$. After these two steps, the remaining Schur complement is the Schur complement of S after eliminating p, tensored with W_2.*

Proof. The first pivot p of $S \otimes W_2$ is indeed a complete pivot because $\|S \otimes W_2\|_{\max} = |p|$. After this elimination, every entry of the resulting Schur complement has magnitude at most $2|p|$, while its upper-left entry is $-p - p^2/p = -2p$. Thus, the second pivot is also a complete pivot.

The two steps together eliminate the upper-left 2×2 block pW_2 of $S \otimes W_2$. Therefore, for $i, j > 1$, the corresponding block of the remaining Schur complement is

$$S_{ij}W_2 - (S_{i1}W_2)(pW_2)^{-1}(S_{1j}W_2) = \left(S_{ij} - \frac{S_{i1}S_{1j}}{p}\right) W_2,$$

the claimed tensor-product Schur complement. □

Using the fact that $W_{2n} = W_n \otimes W_2$, applying this lemma twice gives that any sequence of k complete pivots on S lifts to a sequence of $4k$ complete pivots on $S \otimes W_4$. Moreover, the Schur complement remaining after the lifted sequence is the original remaining Schur complement tensored with W_4. This allows us to lift an initial complete-pivoting sequence on W_n to one on W_{4n}.

The five stages. We start with W_{16} and choose ten complete pivots, leaving a 6×6 Schur complement with entries ± 4. After lifting the sequence to W_{64}, the remaining Schur complement has order 24. We then choose sixteen more pivots so that the resulting 8×8 Schur complement has entries ± 16. Repeating these steps gives the five stages shown in Figure 1. At each stage, the entries of the final Schur complement have magnitude $n/4$, while its order increases by two.

The lifting lemma determines the pivots in each lifted sequence, so only the additional choices made in the smaller Schur complements need to be checked directly. These choices were found computationally by searching among the entries of largest magnitude in the current Schur complement. We specify this pivot selection in Appendix A. We perform the Schur updates in rational arithmetic and verify that each selected pivot has maximum absolute value in the current matrix.

Thus, for $n = 4096$, the first $n - 14 = 4082$ complete pivoting steps leave a Schur complement $(n/4)K$, where K is a particular 14×14 sign matrix. To obtain a pivot exceeding n, it remains to show that complete pivoting on K produces a pivot of magnitude greater than 4.

The pivot exceeding the order. Continuing complete pivoting on $(n/4)K$ gives the final fourteen pivot magnitudes

$$(p_{n-13}, \ldots, p_n) = \frac{n}{4}\left(1, 2, 2, 2, 4, 4, \frac{9}{2}, \frac{10}{9}, \frac{6}{5}, \frac{4}{3}, 1, 2, 2, 4\right). \tag{1}$$

In particular, the seventh pivot in this sequence has magnitude $9n/8$. Explicitly, after six eliminations on $(n/4)K$, we obtain

$$\frac{n}{8}\begin{pmatrix} -4 & 0 & -4 & 0 & 0 & 0 & 0 & 0 \\ 1 & -3 & 3 & 0 & -1 & -2 & 1 & 1 \\ -3 & 1 & -1 & 0 & -1 & 2 & 1 & 1 \\ -2 & 2 & -2 & 0 & 2 & 0 & 2 & -2 \\ -1 & -1 & 1 & 0 & 1 & 2 & -1 & -1 \\ -2 & 2 & -2 & 0 & 2 & 0 & -2 & 2 \\ -4 & 4 & -4 & 4 & 0 & 0 & 0 & 0 \\ 7 & -5 & 9 & -4 & -3 & -2 & -1 & -1 \end{pmatrix}.$$

The maximum absolute entry of this matrix is $9n/8$. It is therefore a valid complete pivot, namely $p_{n-7} = 9n/8$.

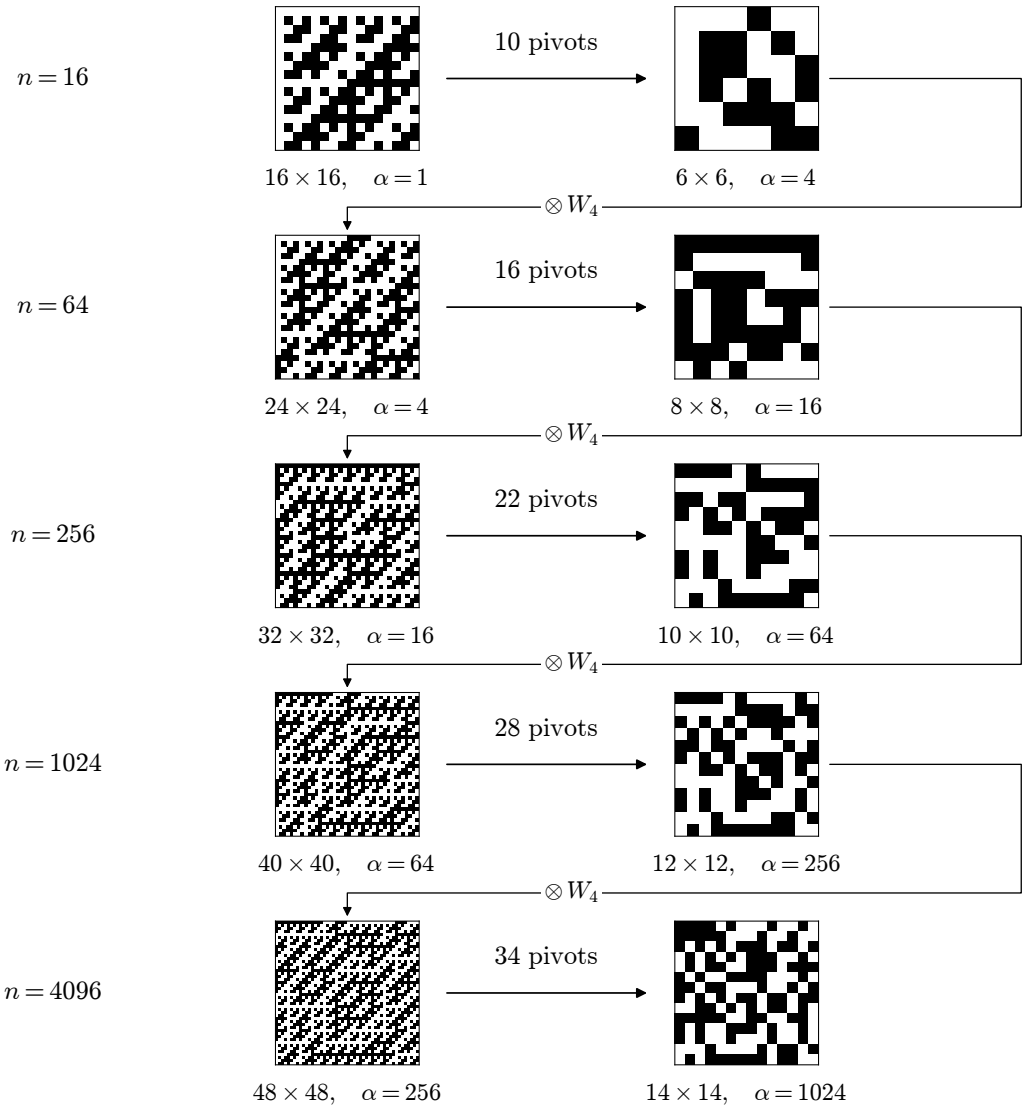


Figure 1: The five stages of the construction. Each matrix shown is a Schur complement divided by its entry magnitude α. Black entries are -1 and white entries are $+1$.

Proof of Theorem 1. The construction above, together with Lemma 1, gives a sequence of complete pivots for W_{4096}. By applying the corresponding row and column permutations in advance, we obtain a completely pivoted matrix $H = PW_{4096}Q$.

Since row and column permutations preserve the orthogonality of W_{4096}, H is Hadamard. The pivots before the final fourteen steps have magnitude at most 1024, while (1) shows that the largest of the remaining pivots is $p_{4089} = 4608$. Hence

$$g(H) = p_{4089} = 4608 = \frac{9}{8}\,4096.$$

□

3 Larger orders and asymptotic growth

The lifting argument also shows that this counterexample extends to all orders 2^d with $d \geq 12$.

Corollary 1. *For every integer $d \geq 12$, there exists a completely pivoted row and column permutation of W_{2^d} whose growth factor is $(9/8)2^d$.*

Proof. Let H be completely pivoted. By Lemma 1, we can choose an ordering of $H \otimes W_2$ that is completely pivoted and satisfies

$$g(H \otimes W_2) = 2g(H).$$

Starting with the matrix H from Theorem 1 and applying this construction repeatedly proves the result. □

We can express this result in terms of the largest possible growth factor among completely pivoted Hadamard matrices of a given order. For any n for which a Hadamard matrix exists, define

$$\Gamma_{\mathrm{Had}}(n) := \max\{g(H) : H \text{ is a CP Hadamard matrix of order } n\}.$$

By Corollary 1, we have

$$\Gamma_{\mathrm{Had}}(2^d) \geq \frac{9}{8}\,2^d, \qquad d \geq 12.$$

Shah and Urschel [7] showed that the largest growth factor under complete pivoting for arbitrary real matrices is $n^{\Theta(\log n)}$. Combined with the restricted-entry result of Edelman and Urschel [4, Theorem 3.3], this implies the same asymptotic behavior for sign matrices. We conjecture that imposing the additional requirement of orthogonality, thereby restricting to Hadamard matrices, does not change this asymptotic behavior.

Conjecture 1. *As $n \to \infty$ through the orders for which Hadamard matrices exist,*

$$\Gamma_{\mathrm{Had}}(n) = n^{\Theta(\log n)}.$$

Although Sylvester's construction gives Hadamard matrices of arbitrarily large order, it remains unknown whether a Hadamard matrix exists at every order divisible by four (the classical Hadamard conjecture [5]). Thus, even the set of orders for which $\Gamma_{\mathrm{Had}}(n)$ is defined is not fully understood.

The upper bound already follows from the general complete pivoting bound. Thus the conjecture asserts that there is a constant $c > 0$ such that $\Gamma_{\mathrm{Had}}(n) \geq n^{c \log n}$ for all sufficiently large Hadamard orders n. Since in our lifting construction the ratio $g(H)/n$ is constant, it does not give this conjectured quasipolynomial lower bound.

Acknowledgements

AI tools were used extensively in constructing this counterexample. In particular, GPT-6 Astra Pro produced the counterexample after a long exploration into this conjecture. Selected prompts and conversations from this exploration, together with scripts for verifying all computational results, are available at `https://github.com/matias-ma/hadamard_cp`. The author organized the final presentation and takes full responsibility for all content.

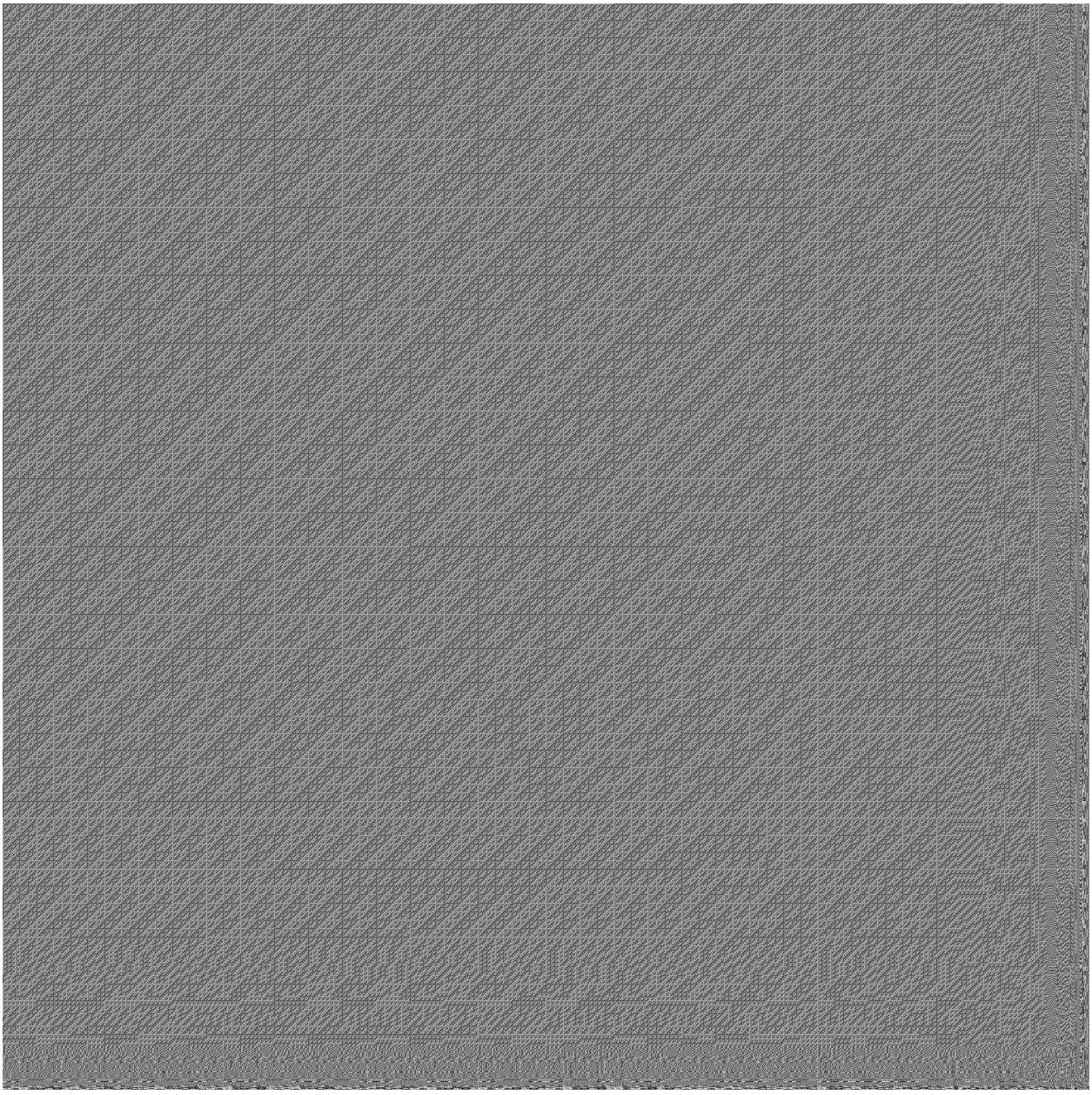


Figure 2: The completely pivoted 4096×4096 Hadamard matrix H from Theorem 1. Black denotes -1 and white denotes $+1$. Each pixel represents one entry.

A Pivot choices

We give the pivot choices used in the five stages of the construction in the sense of Figure 1. Each pair in the lists below gives the row and column of the entry chosen as the next pivot. The positions are zero-indexed in the current Schur complement. For example, the first pair $(14, 8)$ selects row 14 and column 8 of W_{16} and the next pair $(12, 6)$ selects row 12 and column 6 of the remaining 15×15 matrix.

At each step, we check that the selected entry has maximum absolute value in the current matrix. Letting the pivot be $p = S_{ij}$, the remaining entries are updated by

$$S_{ab} \longmapsto S_{ab} - \frac{S_{aj} S_{ib}}{p}, \qquad a \neq i, \quad b \neq j.$$

We then delete row i and column j, keeping the other rows and columns in their existing order, and number the remaining positions from zero again. This ordering is used throughout the lists. Between stages, we lift the previous pivot sequence as in Lemma 1. The remaining Schur complement becomes $S \otimes W_4$. We then continue with the next list.

The lists contain 124 choices in total, all made in Schur complements of order at most 48. Checking them in exact rational arithmetic verifies that every choice is a complete pivot and that each stage ends with the scaled sign matrices illustrated in Figure 1. Together with the lifting lemma, these checks verify all 4096 pivots. The first 4082 pivots have magnitude at most 1024, and the final fourteen have the magnitudes recorded in (1), with the largest being 4608.

Order 16. Starting with W_{16}, choose the following ten pivots, leaving a 6×6 Schur complement with entries ± 4.

$$(14, 8),\ (12, 6),\ (10, 4),\ (9, 2),\ (7, 4),\ (6, 10),\ (8, 6),\ (7, 4),\ (5, 1),\ (3, 1).$$

Order 64. After lifting, the Schur complement has order 24. Choose the following sixteen pivots, leaving an 8×8 Schur complement with entries ± 16.

$$(19, 17),\ (15, 1),\ (11, 8),\ (3, 18),\ (18, 13),\ (6, 4),\ (16, 9),\ (15, 10),$$
$$(0, 0),\ (3, 1),\ (0, 1),\ (1, 2),\ (8, 6),\ (8, 7),\ (6, 9),\ (5, 7).$$

Order 256. After lifting, the Schur complement has order 32. Choose the following twenty-two pivots, leaving a 10×10 Schur complement with entries ± 64.

$$(3, 1),\ (10, 8),\ (13, 15),\ (16, 26),\ (19, 18),\ (22, 11),\ (25, 20),\ (3, 5),\ (3, 3),\ (3, 4),\ (9, 14),$$
$$(9, 19), (11, 10),\ (9, 10),\ (6, 8),\ (12, 6),\ (9, 13),\ (12, 7),\ (12, 10),\ (4, 8),\ (9, 4),\ (5, 10).$$

Order 1024. After lifting, the Schur complement has order 40. Choose the following twenty-eight pivots, leaving a 12×12 Schur complement with entries ± 256.

$$(3, 1),\ (6, 16),\ (9, 35),\ (16, 23),\ (19, 30),\ (22, 26),\ (25, 12),\ (32, 18),\ (28, 4),\ (9, 8),$$
$$(9, 6),\ (9, 7),\ (11, 18),\ (11, 8),\ (8, 1),\ (7, 1),\ (16, 14),\ (21, 22),\ (16, 6),\ (2, 7),\ (12, 7),$$
$$(1, 12),\ (17, 16),\ (11, 3), (14, 2),\ (8, 10),\ (12, 7),\ (8, 5).$$

Order 4096. After lifting, the Schur complement has order 48. Choose the following thirty-four pivots, leaving the 14×14 Schur complement with entries ± 1024.

$$(3, 1),\ (6, 16),\ (9, 4),\ (16, 7),\ (23, 21),\ (26, 24),\ (12, 35),\ (32, 10),\ (35, 16),\ (38, 36),\ (18, 25),\ (24, 29),$$
$$(24, 27),\ (24, 28),\ (17, 13),\ (16, 14),\ (19, 29),\ (3, 26),\ (29, 14),\ (4, 27),\ (18, 25),\ (21, 19),\ (6, 0),\ (2, 9),$$
$$(18, 16),\ (13, 3),\ (4, 5),\ (1, 2),\ (19, 7),\ (16, 0),\ (8, 2),\ (15, 3),\ (10, 7),\ (7, 6).$$

The final fourteen pivots. Continue on this 14×14 Schur complement with the following choices. These give the final pivot magnitudes in (1).

$$(8,4),\ (2,6),\ (10,11),\ (3,8),\ (1,5),\ (2,5),\ (7,2),\ (0,1),\ (0,1),\ (0,0),\ (0,0),\ (0,0),\ (0,0),\ (0,0).$$

Ordering the rows and columns of W_{4096} according to the full elimination sequence, including the lifted pivots, places these complete pivots on the diagonal. The resulting matrix H is therefore completely pivoted, Hadamard, and has growth factor 4608, giving the matrix in Theorem 1.

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